

CoRoT-2b

R-0233

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230807 · created 2026-07-17T03:53:31+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460163.7298 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2196 +/- 0.0085
Transit depth (Rp/Rs)^2	4.82 +/- 0.38 [%]
Orbital Inclination (inc)	87.31 +/- 1.28
Airmass coefficient 1 (a1)	1.258 +/- 0.014
Airmass coefficient 2 (a2)	-0.191 +/- 0.075
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#1 - [211, 200]
Optimal Aperture	4.01
Optimal Annulus	17.12
Transit Duration (day)	0.0531 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.2196 $\pm$ 0.0085 vs 0.1667 (deviation 6.06 σ)
Duration fit vs published	0.0531 d vs 0.095 d (deviation 15.11 σ)
Mid-time O-C	5.1 min
Depth SNR	25.2
Residual scatter	1.12 %

Light curve

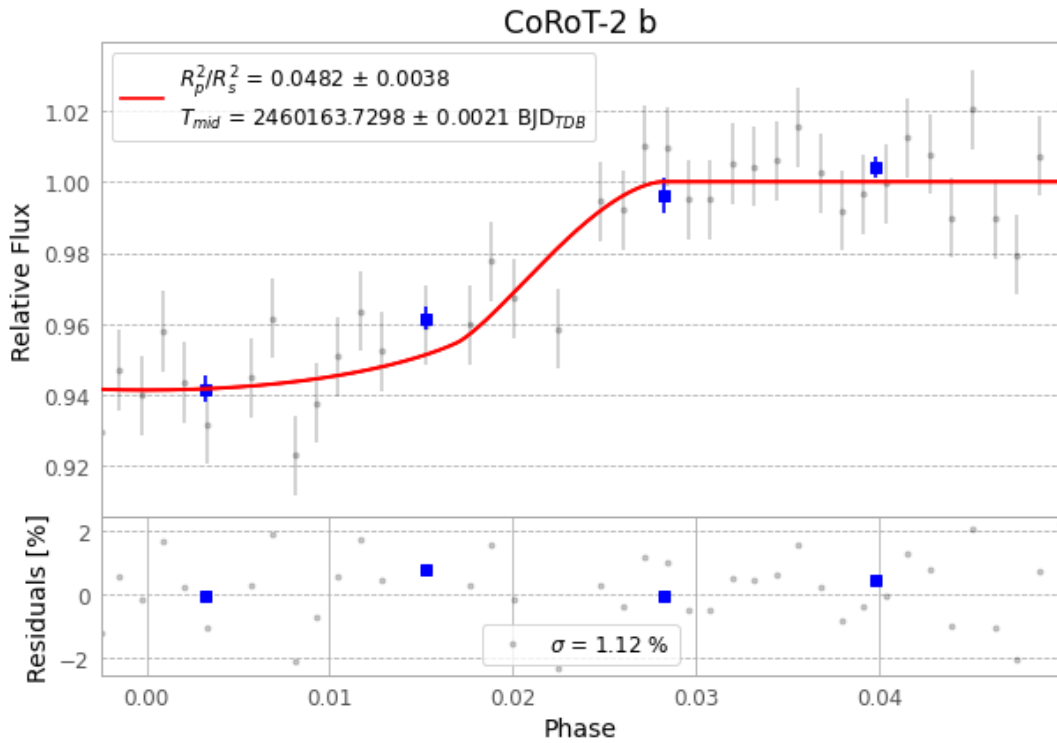


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-08-06.png (SHA-256 dd30ae32e05b3e66...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c2c27e0b0e3ec6be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

40 FITS frames (26 MB), CoRoT-2230807051546.FITS → CoRoT-2230807072715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230807.log (c5b54e19e66b...), FinalLightCurve_CoRoT-2 b_2023-08-06.png (dd30ae32e05b...), FinalLightCurve_CoRoT-2 b_2023-08-06.csv (60c6e926f8df...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 03:53Z.